

Article

Anti-Wind Disturbance Algorithms for Small Rotorcraft UAVs

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Abstract

Small rotorcraft unmanned aerial vehicles (UAVs) are highly susceptible to wind disturbances when performing tasks such as fixed-point hovering, low-altitude inspection, and aggressive maneuvers. Under complex, variable meteorological conditions, attitude stability and position-holding accuracy are particularly critical. Although quadrotor UAVs exhibit structural and dynamic symmetry, real wind disturbances are often asymmetric, disrupting the original balance and leading to intensified attitude oscillations, position drift, and degraded data quality. To effectively address the challenges of wind-induced oscillation and positional deviation, this paper proposes a fuzzy logic-based linear active disturbance rejection control (Fuzzy-LADRC) strategy. This approach employs a hybrid algorithm combining particle swarm optimization and gray wolf optimization to optimize controller parameters and incorporates fuzzy logic to enhance the adaptive capability of the linear active disturbance rejection controller (LADRC). Simulation experiments conducted in MATLAB/Simulink under complex wind-field conditions demonstrate that the proposed method significantly outperforms traditional PID controllers: in the regulation of roll and pitch angles, control performance improves by approximately 5%, while in yaw angle control, the improvement reaches up to 30%. Furthermore, this method can significantly suppress position deviation and fluctuation in the X and Y directions, and reduce the overshoot in the Z-axis during the UAV's takeoff phase by 75%.

Keywords: small rotorcraft UAV; wind disturbance rejection; Fuzzy-LADRC; parameter tuning



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1. Introduction

Turbulent gusts significantly oscillate unmanned aerial vehicles (UAVs), causing deviations from intended flight paths and, in severe cases, loss of control. This issue is critical in UAV-based inspection missions that require single-point data acquisition or repeated imaging at specific monitoring sites. Camera platform instability and positional offsets degrade data quality and reduce the reliability of subsequent analysis. Advanced UAV control algorithms must rapidly detect and precisely compensate for asymmetric wind disturbances to ensure data quality and mission success.

The current control algorithms for UAVs fall into three categories: linear, nonlinear, and intelligent. Linear control algorithms, such as proportional–integral–derivative (PID) [1] and linear quadratic regulator (LQR) [2], are characterized by their straightforward implementation and inherent stability. This has led to their widespread adoption in industrial control applications. However, their performance is often compromised in the presence of nonlinear system dynamics or significant disturbances. Nonlinear control algorithms, such as backstepping and active disturbance rejection control (ADRC) [3], address the limitations of linear control. These algorithms require minimal precision from the system model and can suppress external disturbances to some extent. However, they have many control parameters, and computation and tuning are complex. This complexity can result in suboptimal performance in complex systems. Intelligent control algorithms, such as neural networks and fuzzy control, offer certain advantages compared to the other two approaches. These algorithms are highly adaptive and resistant to external disturbances. However, they are computationally intensive and difficult to interpret. The three primary challenges that must be addressed are inadequate control under disturbance, complexity of parameter tuning, and limited adaptive capabilities.

To address the suboptimal control performance of linear control algorithms, these algorithms could be optimized. CAN and Ercan [4] proposed a PID controller based on particle swarm optimization and differential evolution. Sun and Wan proposed a BPPID-LADRC controller [5] that integrates the LADRC algorithm with PID control. Building on this, a multi-loop LQR controller with parameter tuning implemented via an improved genetic algorithm was introduced in [6].

Subsequently, addressing the challenge of intricate parameter tuning in nonlinear controllers, researchers have refined nonlinear control algorithms. Specifically, Cui et al. introduced a fault-tolerant control algorithm that integrates control allocation principles with backstepping techniques [7]. Yan et al. have designed a fractional-order improved PID controller by integrating feedforward compensation with the concept of incomplete differentiation [8]. XU et al. integrated an ADRC with a PID controller for attitude control [9], subsequently incorporating high-gain parameters into a backstepping controller within the position loop. While these optimization algorithms have demonstrated superior performance compared to baseline control algorithms, parameter tuning remains complicated.

With the advancements in computer science, neural networks and fuzzy control have been increasingly applied to flight control systems. Neural networks can identify optimal solutions through continuous data learning and iterative optimization, thereby enabling effective flight control. Furthermore, algorithm optimization can be achieved by integrating neural networks with other control algorithms to enhance the system's control capabilities. Examples include adaptive control and PID control algorithms. Fuzzy control transforms precise mathematical computations into fuzzy-logic reasoning, employing a feedback mechanism to adjust based on the system's actual conditions, thereby enabling adaptive control. Compared to neural networks, fuzzy control systems offer enhanced interpretability, as well as robustness, stability, and adaptability. Consequently, to address the limitations in the adaptive capability of controllers, numerous researchers have integrated fuzzy control with other control algorithms [10]. For instance, Tran et al. integrated fuzzy control with adaptive algorithms to develop a type-2 fuzzy adaptive controller that ensures stable flight of a quadrotor UAV under varying payloads. Kayacan integrated fuzzy control with neural networks to design a type-2 fuzzy neural network (T2FNN) [11], which demonstrated robust performance under wind gusts. Fuzzy control and backstepping techniques were integrated to develop an adaptive type-2 fuzzy backstepping controller [12], which enabled parameter adaptation within the backstepping controller, thereby enhancing overall control

performance. In recent years, adaptive and compliant control strategies have emerged as a prominent research paradigm for robotic systems operating in uncertain environments. It has been widely reported that integrating adaptive mechanisms with compliant control structures can significantly improve system stability and control performance under load disturbances [13].

Reducing parameter tuning time and enhancing adaptive control improve a UAV's wind disturbance rejection. This paper proposes a Fuzzy-LADRC method tuned by a hybrid Particle Swarm Optimization–Gray Wolf Optimization (PSO-GWO) algorithm that combines particle swarm optimization and gray wolf optimization to efficiently tune controller parameters. Fuzzy control further optimizes LADRC parameters, enhancing adaptive capability under wind disturbance. A quadrotor UAV model and wind simulation in MATLAB/Simulink 2022a validate the Fuzzy-LADRC method and parameter tuning with the hybrid PSO-GWO algorithm.

The main contributions of this work are threefold: (1) establishing a Fuzzy-LADRC (Fuzzy-enabled Linear Active Disturbance Rejection Controller) framework for wind-disturbance rejection in small rotorcraft UAVs; (2) introducing a hybrid PSO-GWO (Particle Swarm Optimization–Gray Wolf Optimization) method for tuning the key parameters of the LADRC; (3) providing quantitative validation under multiple wind-field conditions.

2. Fuzzy-LADRC Design

The control algorithm is the core of the flight control system and directly determines the control performance of quadrotor UAVs. The LADRC introduces the concept of bandwidth parameterization, which significantly reduces the number of tuning parameters and simplifies the tuning process. Meanwhile, fuzzy control enables adaptive adjustment of controller parameters. By integrating the advantages of both approaches, this paper proposes a Fuzzy-LADRC control method.

2.1. Quadrotor UAV Model Under Wind Disturbance

To support the controller design, a simplified quadrotor UAV model under wind disturbance is adopted. The UAV is modeled as a rigid body in the inertial and body-fixed frames, with (x, y, z) as the position variables and (ϕ, θ, ψ) as the attitude variables. The translational and rotational dynamics are described by the Newton–Euler method, while wind effects and modeling uncertainty are treated as lumped disturbances. For control implementation, the altitude and attitude channels are decoupled and approximated as second-order subsystems with total disturbance, which provides the plant basis for the proposed Fuzzy-LADRC method.

2.2. Structure and Principle of LADRC Controller

The LADRC mainly consists of a linear extended state observer (LESO) and a linear state error feedback (LSEF) control law. Its basic structure is illustrated in Figure 1. The LESO can estimate both external disturbances and internal uncertainties of the system in real time, while the LSEF compensates for these disturbances to ensure stable system operation. The integration of LESO and LSEF enables more accurate control and improves system performance and stability.

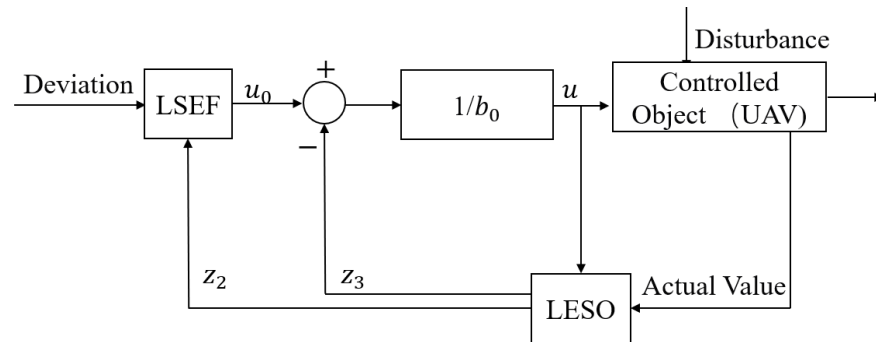


Figure 1. LADRC structure diagram.

The LESO can estimate the system states z_1, z_2 , as well as the total disturbance z_3 . The LESO is expressed as

$$\begin{cases} \dot{Z}_1 = Z_2 - k_1(Z_1 - y) \\ \dot{Z}_2 = Z_3 - k_2(Z_1 - y) + bu \\ \dot{Z}_3 = -k_3(Z_1 - y) \end{cases} \quad (1)$$

where z_1 and z_2 denote the estimated system states, z_3 represents the estimated total disturbance, k_1, k_2, k_3 are the observer gains, y is the system output, u is the control input, and b is the control gain.

The LSEF computes the intermediate control signal u_0 based on the estimated states and then determines the final control input by compensating for the estimated disturbance. The control law is given by

$$\begin{cases} u_0 = k_p(v - Z_1) + k_d Z_2 \\ u = \frac{u_0 - Z_3}{b_0} \end{cases} \quad (2)$$

where k_p and k_d are the proportional and derivative gains, respectively, v is the reference input, u_0 is the intermediate control signal, and b_0 is the estimated control gain used for compensation.

By introducing the observer bandwidth ω_0 and the controller bandwidth ω_c , the parameters of the LADRC can be systematically simplified. Specifically, by designing the observer error dynamics to have the standard characteristic polynomial $(s + \omega_0)^3$, the observer gains can be expressed as $k_1 = 3\omega_0$, $k_2 = 3\omega_0^2$, $k_3 = \omega_0^3$. Similarly, by shaping the controller error dynamics as a standard second-order system with a characteristic polynomial $(s^2 + 2\omega_c s + \omega_c^2)$, the controller parameters can be determined as $k_p = \omega_c^2$, $k_d = 2\omega_c$.

This bandwidth-based parameterization reduces the number of tuning parameters and provides clear physical interpretations. Since the exact value of b is generally unknown, it is replaced by an estimated value b_0 . This design satisfies the separation principle, ensuring the independence of observer and controller design while maintaining system stability. Overall, the LADRC framework transforms an uncertain system into an approximately disturbance-free system, thereby significantly enhancing the disturbance rejection capability.

2.3. Fuzzy Control Theory

Fuzzy control, an algorithm rooted in human control cognition, leverages human understanding of the controlled system [14]. It partitions the membership functions of fuzzy variables based on empirical knowledge and infers control outputs based on the controlled system's feedback state. Fuzzy control systems do not require precise mathematical models of the controlled object during operation; instead, they can continuously infer and control the object based solely on its state feedback.

The fuzzy controller primarily functions through three core stages: fuzzification, fuzzy inference, and defuzzification (Figure 2):

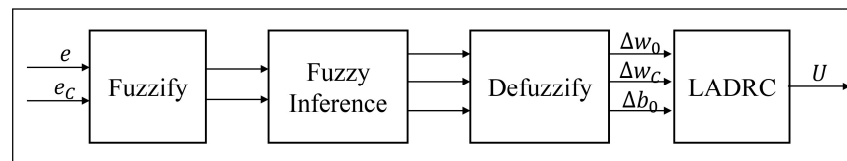


Figure 2. Fuzzy control process.

(1) Fuzzify

The fuzzification process transforms input data into states suitable for the fuzzy inference rules. In practical applications, the raw input signals are not directly usable for fuzzy rule inference. Therefore, the input signals' base universe of discourse must be mapped into the fuzzy domain.

(2) Fuzzy Inference

Fuzzy inference relies on the construction of fuzzy rules. Within these rules, the fuzzy domain of variables is expressed using fuzzy linguistic terms. This approach provides a clearer representation of a variable's state within its domain, thereby facilitating the formulation of effective fuzzy rules.

(3) Defuzzify

Following the fuzzification of the input signal into a fuzzy linguistic variable, the output of the fuzzy inference process remains a fuzzy linguistic variable. Since fuzzy linguistic variables are not directly applicable for control purposes, it is necessary to defuzzify the inferred fuzzy linguistic variable into a crisp control signal that can be recognized by the controlled system.

2.4. Fuzzy-LADRC Controller Design

The design of a fuzzy controller requires determining the fuzzy dimension, i.e., the number of input variables, based on the characteristics of the controlled system. Furthermore, the fundamental universes of discourse and fuzzy universes must be established for both input and output variables, encompassing the initial and rate-of-change variables. Finally, the fuzzy rules governing the inference of the output fuzzy universe from the input fuzzy universe must be defined.

Considering the characteristics of the LADRC, a two-dimensional fuzzy controller is employed. The quantitative deviation, denoted by e , and the rate of change in the quantitative deviation e_c serve as inputs. These inputs are quantized via scaling factors and subsequently utilized as inputs for the fuzzy self-tuning LADRC. The outputs, representing the adjustment values for the parameters w_0 , w_c and b_0 , are denoted by V_{w_0} , V_{w_c} and V_{b_0} , respectively.

The outputs correspond to the adjustment values of the three key LADRC parameters, namely the observer bandwidth w_0 , the controller bandwidth w_c , and the system gain b_0 . w_c denotes the controller bandwidth, which governs the dynamic response of the closed-loop system. A proper increase in w_c can accelerate the response, but an excessively large value may impair system stability. w_0 denotes the observer bandwidth, which determines the tracking speed of the state observer. Increasing w_0 within a suitable range improves the observer performance, whereas an overly large value may lead to excessive tracking oscillation and reduced observation accuracy. b_0 denotes the control input compensation coefficient, which is obtained from the initial system input.

The overall control structure is illustrated in Figure 3. The desired value is compared with the actual value to generate the error signal. The fuzzy controller takes the error and its

rate of change as inputs to adaptively tune the LADRC parameters. The LSEF generates the control input, while the LESO estimates the total disturbance in real time for compensation. The plant represents the UAV system under external disturbances, forming a complete closed-loop control system.

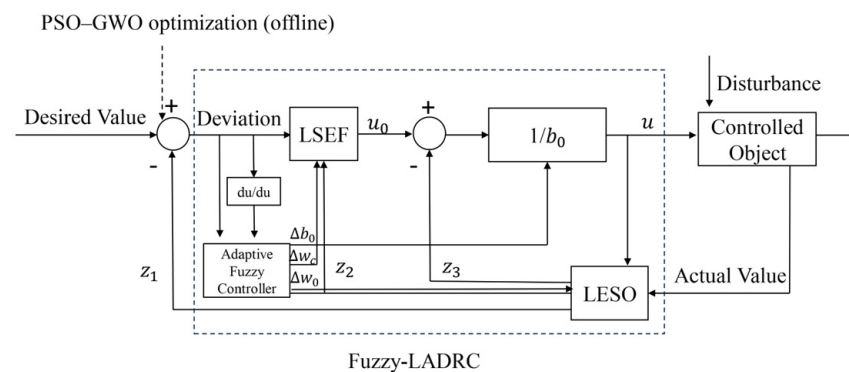


Figure 3. Fuzzy-LADRC structure diagram.

(1) Selection of Fuzzy Sets and Scaling Factors for Fuzzy Domain Quantization.

The fuzzy linguistic variables are categorized based on their fuzzy domain, positive, negative, or zero, based on the polarity of their values. Alternatively, they can also be classified as large, medium, or small based on the magnitude of their values. A more detailed partitioning of the fuzzy domain results in more complex fuzzy rules and slower inference speeds.

In this study, the fuzzy sets for the input variables, e and e_c , are partitioned into seven levels, respectively. Specifically, the fuzzy sets for the quantitative deviation, e , are defined as {NB, NM, NS, ZO, PS, PM, PB}. NB, NM, and NS denote the negative input deviations, indicating that the system's actual value exceeds the desired value. These terms represent the quantitative deviations: negative large, negative medium, and negative small, respectively, indicating that the actual value is significantly higher, moderately higher, or slightly higher than the desired value. ZO represents the condition where the actual value equals the desired value. PS, PM, and PB denote the positive input deviations, indicating that the actual value is below the desired value. These three fuzzy linguistic terms represent positive small, positive medium, and positive large, respectively, indicating that the actual value is slightly lower, moderately lower, and significantly lower than the desired value.

The fuzzy sets for the quantitative deviation rate, denoted as e_c , are defined as {NB, NM, NS, ZO, PS, PM, PB}. Here, NB, NM, and NS represent the trend of the actual value increasing relative to the desired value, indicating rapid, moderate, and slow increases, respectively. ZO indicates that the actual value changes synchronously with or remains unchanged from the desired value. PS, PM, and PB indicate a decreasing trend in the actual value relative to the desired value, with PS, PM, and PB representing slight, moderate, and rapid decreases in the input quantity, respectively.

The output fuzzy variable set is partitioned into seven levels, where {NB, NM, NS, ZO, PS, PM, PB} denote the output adjustment states: Negative Big, Negative Medium, Negative Small, Zero, Positive Small, Positive Medium, and Positive Big, respectively. The fuzzy rule tables are presented in Tables 1–3.

Table 1. Fuzzy rule table for V_{w_0} .

V_{w_0}		e						
		NB	NM	NS	ZO	PS	PM	PB
e_c	NB	PB	PB	PM	PM	PS	ZO	ZO
	NM	PB	PB	PM	PM	PS	ZO	ZO
	NS	PM	PM	PS	PS	PS	PS	NS
	ZO	PM	PS	PS	NB	NB	NB	NS
	PS	NS	NS	NS	NS	NS	NM	NM
	PM	PM	PM	PB	PB	PB	PB	PB
	PB	PB	PB	PB	PB	PB	PB	PB

Table 2. Fuzzy rule table for V_{w_c} .

V_{w_c}		e						
		NB	NM	NS	ZO	PS	PM	PB
e_c	NB	PB	PB	PM	PM	PS	ZO	ZO
	NM	PB	PB	PM	PM	PS	ZO	ZO
	NS	PM	PM	PS	PS	ZO	ZO	NS
	ZO	PM	PS	PS	ZO	NS	NS	NM
	PS	PS	ZO	ZO	NS	NM	NM	NM
	PM	ZO	ZO	NS	NM	NM	NB	NB
	PB	ZO	NS	NM	NM	NB	NB	NB

Table 3. Fuzzy rule table for V_{b_0} .

V_{b_0}		e						
		NB	NM	NS	ZO	PS	PM	PB
e_c	NB	NB	NB	NM	NM	NS	ZO	ZO
	NM	NB	NB	NM	NM	NS	ZO	ZO
	NS	NM	NM	NS	NS	ZO	ZO	PS
	ZO	NM	NS	NS	ZO	PS	PS	PM
	PS	NS	NS	ZO	PS	PM	PM	PB
	PM	ZO	ZO	PS	PM	PM	PB	PB
	PB	ZO	PS	PM	PM	PB	PB	PB

(2) Selection of Membership Functions

The choice of membership functions depends on the input variables, the desired precision and sensitivity, and the control rules. Triangular membership functions are often preferred due to their computational efficiency and low memory requirements. They also offer sensitivity to error variations and a uniform distribution. We have opted to use triangular functions as the membership functions for both input and output variables [15]. Furthermore, to ensure ease of membership function design while adhering to the domain partitioning principle, the universe of discourse for each fuzzy subset is partitioned using a uniform division method.

The fuzzy controller fuzzifies the error and its rate of change, then infers based on the established fuzzy rules, thereby adaptively generating real-time correction values for the LADRC parameters. Accordingly, the variations in the three LADRC parameters w_0 , w_c , and b_0 are denoted by V_{w_0} , V_{w_c} , and V_{b_0} , respectively. The parameter tuning process is described as follows:

$$\begin{cases} b_0 = \hat{b}_0 + \Delta b_0 \\ w_0 = \hat{w}_0 + \Delta w_0 \\ w_c = \hat{w}_c + \Delta w_c \end{cases} \quad (3)$$

The surface plots of the fuzzy controller outputs V_{w_0} , V_{w_c} and V_{b_0} , along with the membership functions for e and e_c , are presented in Figures 4–7.

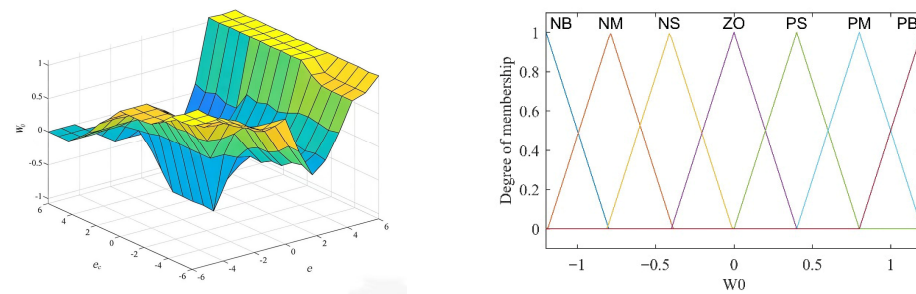


Figure 4. Output surface plot and membership function plots for V_{w_0} .

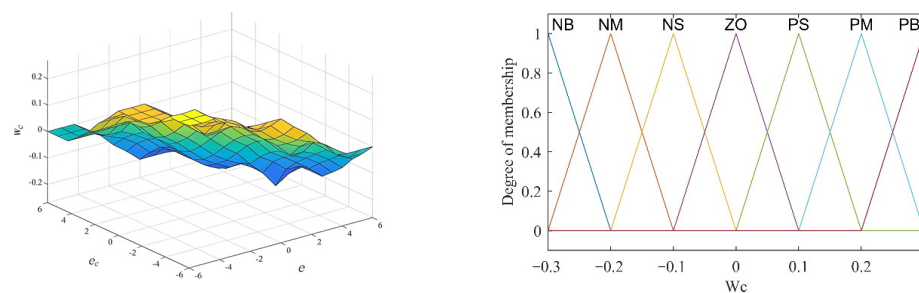


Figure 5. Output surface plot and membership function plots for V_{w_c} .

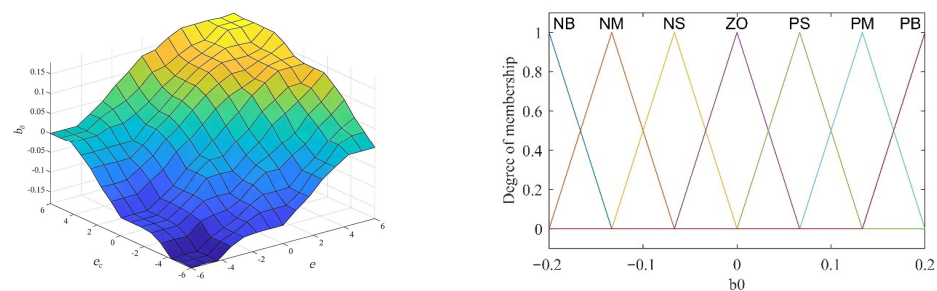


Figure 6. Output surface plot and membership function plots for V_{b_0} .

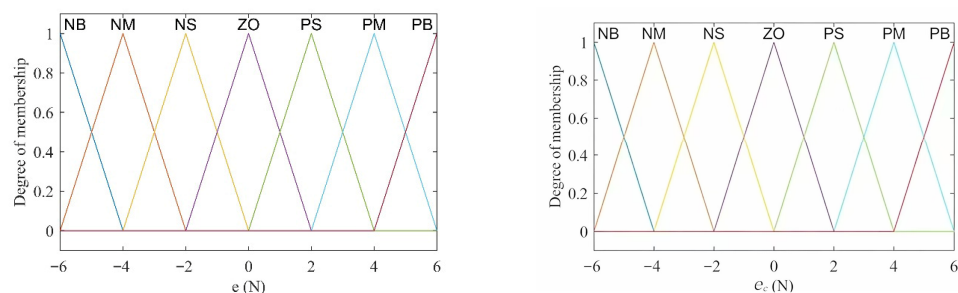


Figure 7. Membership function plots for e and e_c .

Figures 4–7 illustrate the design and adaptive behavior of the proposed fuzzy controller. Figures 4–6 present the output surfaces and membership functions of the LADRC parameters V_{w_0} , V_{w_c} , and V_{b_0} , revealing the nonlinear mapping between the inputs (error e and its rate of change e_c) and the parameter adjustments. The output membership functions determine the magnitude and direction of tuning, ensuring smooth and gradual regulation. Figure 7 shows the membership functions of e and e_c , which characterize the system state by capturing both the magnitude of deviation and its trend. Overall, the results demon-

strate that the proposed fuzzy controller enables adaptive and smooth parameter tuning, achieving a balance between fast response and system stability.

3. Hybrid Particle Swarm Optimization–Gray Wolf Optimization Algorithm

Both the particle swarm optimization (PSO) and gray wolf optimizer (GWO) are swarm intelligence-based optimization algorithms rooted in the observation and imitation of real-world animal populations. To address the limitations and leverage the strengths of both optimization algorithms, we propose a hybrid parameter-tuning method based on a coupled PSO-GWO approach. Using the ITAE performance index as the optimization objective, this method enhances dynamic performance and disturbance rejection while ensuring system stability.

3.1. Particle Swarm Optimization Algorithm

As a heuristic algorithm designed for population optimization problems, PSO draws inspiration from the foraging behavior of bird flocks, employing their search strategies to address optimization challenges. The iterative process of updating particle positions and velocities in PSO, based on individual and global best positions, can mathematically be represented as

$$\omega v_{id}^{k+1} = \omega v_{id}^k + c_1 r_1 (pbest_{id}^k - x_{id}^k) + c_2 r_2 (gbest_d^k - x_{id}^k), \quad (4)$$

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1}, \quad (5)$$

where $i = A$ particle within the particle swarm with the swarm's cardinality of N ; c_1, c_2 = Learning factor; r_1, r_2 = Random values within the interval $[0, 1]$; ω = Inertia weight; x_{id}^k = The velocity of the i th particle is denoted by v_{id}^k after k iterations within the particle swarm; $pbest_{id}^k$ = The optimal position of the i th particle at the k th iteration of the particle swarm; $gbest_{id}^k$ = At the k th iteration of the particle swarm, the globally best position among all particles is determined, and $P_g = (p_{g1}, p_{g2}, \dots, p_{gD})$, the optimal position that the population has reached in each iteration.

3.2. Gray Wolf Optimization Algorithm

The GWO [16] is a metaheuristic algorithm inspired by the hunting behavior of gray wolves. It simulates the wolves' cooperative hunting strategy, particularly the encircling of prey, to achieve optimization goals. The Gray Wolf Optimizer algorithm has been widely adopted for parameter tuning [17–20], disaster identification [21–23], and other domains due to its straightforward structure and ease of parameter configuration. Wind-induced vibration systems are inherently nonlinear and subject to external disturbances. Meanwhile, the proposed fuzzy LADRC requires simultaneous tuning of multiple parameters under time-varying conditions. To address this, the GWO is adopted due to its strong global search capability and simple parameter structure. Compared with conventional optimization methods such as PSO and Genetic Algorithms (GAs), GWO involves fewer control parameters and exhibits more stable convergence, making it well-suited for iterative parameter tuning in nonlinear systems.

3.3. Hybrid Particle Swarm Optimization–Gray Wolf Optimization Algorithm

During the iterative process of the GWO algorithm, the positions of the gray wolves are primarily updated based on the alpha wolf's position, leading to a single-position update strategy and a tendency to converge to local optima. In contrast, the PSO algorithm maintains a record of each particle's historical best position, and its parameters are

straightforward and easily adjustable. Consequently, a hybrid PSO-GWO is proposed to enhance the optimization capability and convergence speed. The particle position update mechanism replaces the gray wolf individual position update, enabling the algorithm to retain the best position information throughout the iterative process. This allows the gray wolf to consider both the leader wolf's position and the individual's historical optimal adaptive position. Furthermore, the inertia weight, denoted by w , is adjusted to balance the hybrid algorithm's global search and local exploitation capabilities. The integrated velocity and position updates are presented as follows:

$$v_{i,j}^{t+1} = w(X_{i,j}^t - x_{i,j}^t) + c_1 r_1 (p_{best}^t - x_{i,j}^t) + c_2 r_2 (g_{best}^t - x_{i,j}^t) \quad (6)$$

$$x_{i,j}^{t+1} = x_{i,j}^t + v_{i,j}^{t+1} \quad (7)$$

The PSO-GWO algorithm is used for offline optimization of the nominal LADRC parameters $\hat{w}_0$, $\hat{w}_c$, and $\hat{b}_0$, whereas the fuzzy tuner performs online adaptive correction through Δw_0 , Δw_c , and Δb_0 .

To quantitatively evaluate the control system's performance and guide the optimization process, a fitness function is constructed to accurately reflect the system's response speed, accuracy, and stability. The integral of time-weighted absolute error (ITAE) is adopted as the performance index, which is defined as

$$ITAE = \int_0^T t |e(t)| dt \quad (8)$$

where $e(t) = r(t) - y(t)$ denotes the tracking error, $r(t)$ is the reference input, $y(t)$ is the actual system output, and T represents the simulation duration.

The ITAE function, defined as the product of time and the absolute error, can dynamically reflect multiple aspects of control performance, including response speed, overshoot, and steady-state error. By minimizing this index, optimal controller parameters can be obtained, thereby improving control accuracy and dynamic performance.

4. Simulation Experiments and Analysis Under Wind Disturbance

In this study, we implement a Fuzzy-LADRC position and attitude controller. The controller incorporates the proposed PSO-GWO tuning method. We use MATLAB/Simulink for implementation, along with a quadrotor UAV model and a wind-field simulation module. Initially, focusing on the altitude control channel, we compared the control data of Fuzzy-LADRC and LADRC to validate the effectiveness of Fuzzy logic in enhancing LADRC performance. Subsequently, we conducted a comparative analysis of LADRC parameter tuning results using PSO and PSO-GWO, thereby verifying the superiority of PSO-GWO. Finally, under various wind disturbance scenarios, we compared the performance of the proposed controller with a traditional cascaded PID controller to demonstrate the advantages of the proposed control strategy.

4.1. Wind Farm Simulation

Based on the principles of wind farm modeling, a Simulink simulation model of the wind farm is constructed, with simulated wind speeds for gusts, stochastic winds, and ramped winds as illustrated in Figure 8.

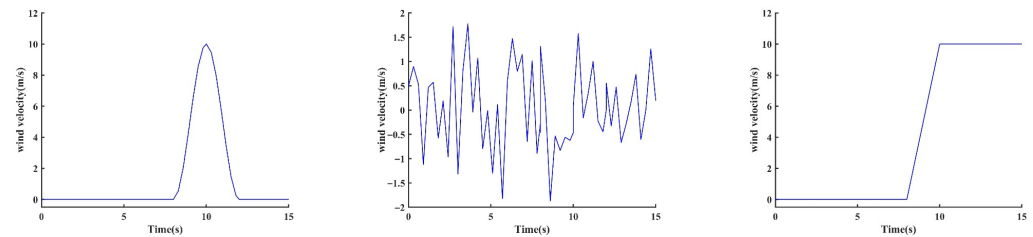


Figure 8. Simulated wind speeds for gust, stochastic, and gradient wind profiles.

4.2. Fuzzy-LADRC

Based on the fuzzy control rules and the domain established previously, a Fuzzy-LADRC controller is constructed. For example, the LADRC controller for the altitude channel dynamically adjusts its gains based on the altitude control channel error and its rate of change. The desired altitude for the Z-axis is set to 2 m, and a disturbance with an amplitude of 1 is added at the 6th second. The simulation results are shown in Figure 9.

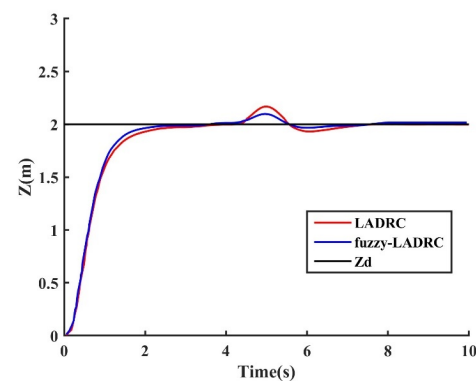


Figure 9. Fuzzy-LADRC altitude control.

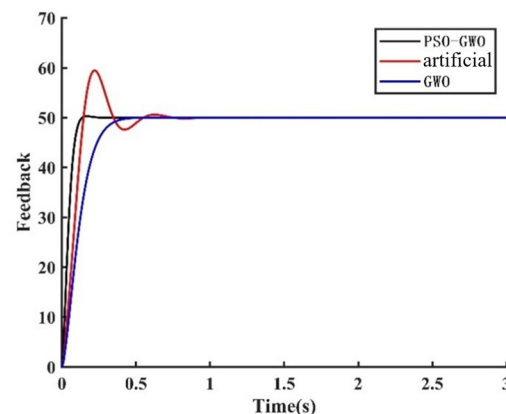
Analysis of the figures indicates that, compared to LADRC, Fuzzy-LADRC achieves more rapid altitude stabilization in quadrotor UAVs. Furthermore, upon the introduction of disturbances, Fuzzy-LADRC facilitates a quicker attainment of stable altitude, accompanied by a 10% reduction in overshoot. The performance metrics of both controllers, derived from simulation experiments, are presented in Table 4. A comparative analysis of the data reveals that Fuzzy-LADRC exhibits a rise time of 3.2 s, which is 0.2 s faster than LADRC, thereby enabling a swifter attainment of the target altitude and demonstrating a superior dynamic response capability, particularly in complex environments. The overshoot of Fuzzy-LADRC is 5%, significantly lower than LADRC's 15%, indicating the absence of substantial oscillations and enabling precise control of altitude variations, thereby ensuring smoother UAV operation. The dynamic settling of Fuzzy-LADRC is recorded at 2%, considerably less than LADRC's 5%, suggesting that the former maintains higher precision during altitude changes and remains stable under complex disturbances. The recovery time for Fuzzy-LADRC is 3.2 s, which is 0.8 s shorter than LADRC, indicating that Fuzzy-LADRC can recover to a stable state more rapidly after disturbances, thereby exhibiting enhanced anti-interference capabilities. In conclusion, the Fuzzy-LADRC, incorporating fuzzy rules, demonstrates superior performance compared to LADRC across response capability, stability, precision control, and anti-interference ability.

Table 4. Comparison of dynamic performance indices of the Fuzzy-LADRC and PID controllers under step input.

Performance Index	Fuzzy-LADRC	LADRC
Rise Time (s)	3.2	3.4
Overshoot	5%	15%
Settling time (s)	3	3.2
Maximum drop after disturbance (%)	2%	5%
Recovery Time (s)	3.2	4

4.3. PSO-GWO

The PSO-GWO and GWO algorithms proposed in this paper were separately applied to the LADRC for parameter tuning. Furthermore, a comparative analysis was conducted between the manual parameter tuning method and the two optimization algorithms. The parameter-optimization response curves for the three methods are shown in Figure 10, and the optimized parameters are listed in Table 5.

**Figure 10.** Parameter optimization response curves.**Table 5.** Parameter optimization.

Method	w_0	w_c	b_0	Fitness Value
Manual tuning	400	20	200	0.42
GWO	642.25	13.45	178.67	0.02
PSO-GWO	567.45	23.66	247.84	0.0005

Based on the results, the PSO-GWO algorithm yields a superior final best fitness value compared to GWO, indicating enhanced global search capability and reduced susceptibility to local optima. Furthermore, the controller employing PSO-GWO exhibits a faster response and, despite a slight overshoot, achieves the shortest settling time, thereby demonstrating optimal overall performance. Manual parameter tuning, which relies on iterative adjustments based on empirical experience, is significantly less efficient than intelligent optimization algorithms.

4.4. Fuzzy-LADRC Tuned by PSO-GWO

Based on the principles of the wind field model, a Simulink simulation model of the wind field is constructed. The anti-wind-disturbance performance of the proposed controller and a conventional cascade PID controller is comparatively analyzed under gust, gradient, and composite wind-field conditions. In all three wind fields, the quadrotor UAV initiates from the origin $[X,Y,Z] = [0,0,0]$ of the Earth coordinate system, ascending along the

positive direction of the Earth coordinate system's X-axis for 2 m. The wind field direction is set at $\pm 20^\circ$, and the simulation commences when the UAV reaches the hovering position.

4.4.1. Simulation and Analysis of Controller Anti-Wind Disturbance Under Gust Conditions

The disturbance begins at 5 s, persists for 10 s, and has a peak wind speed of 10 m/s. The performance comparison between the proposed controller and the conventional cascade PID controller is shown in Figure 11.

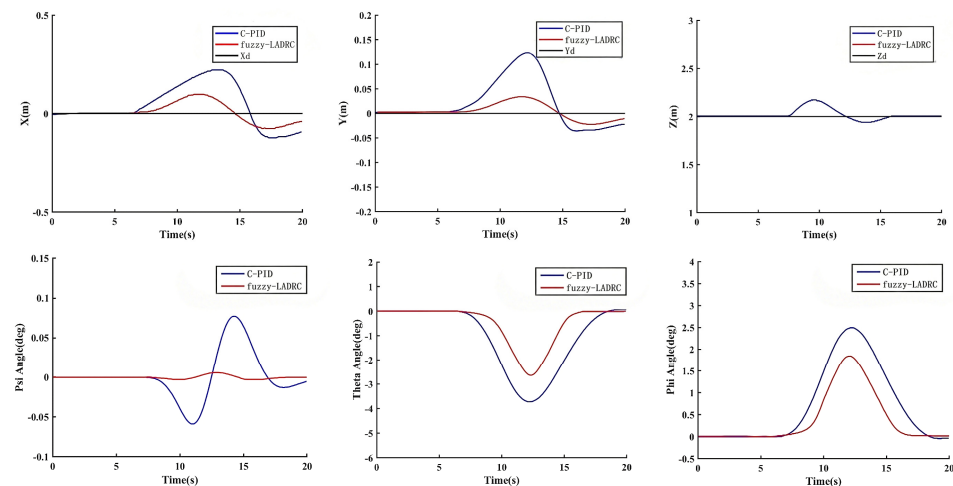


Figure 11. Six-degree-of-freedom response curves of a quadrotor UAV under wind gusts.

The comparison demonstrates that the proposed controller outperforms the PID controller under wind gust disturbances. The response curves of the proposed controller exhibit smoother behavior across all six channels. Notably, the Z-axis and Psi angle responses show minimal oscillations, whereas the PID controller exhibits significant fluctuations in the Psi response, with a settling time of up to 20 s. This indicates that the proposed controller offers superior response speed and stability compared to the conventional cascaded PID controller. The peak fluctuations in pitch and roll angles for the proposed controller are approximately 1.8° and -3° , respectively, compared to 2.6° and -3.7° for the PID controller, with a shorter settling time. In the yaw angle control channel, the proposed controller exhibits negligible oscillations, whereas the cascaded PID controller produces peaks of -0.06° and 0.09° at 11 s and 14 s, respectively. Furthermore, in the position control of the X, Y, and Z axes, the response curves of the proposed controller are closer to zero than those of the PID controller, indicating higher control accuracy. Specifically, the Z-axis position control with the proposed controller shows almost no oscillations, while the PID controller exhibits significant fluctuations.

4.4.2. Simulation and Analysis of Wind Disturbance Rejection Control for Gradient Wind Fields

The ramp-up wind profile initiates at 5 s, with a transition duration of 5 s and a peak wind speed of 10 m/s. The control performance comparison between the proposed controller and the conventional cascade PID controller is presented in Figure 12.

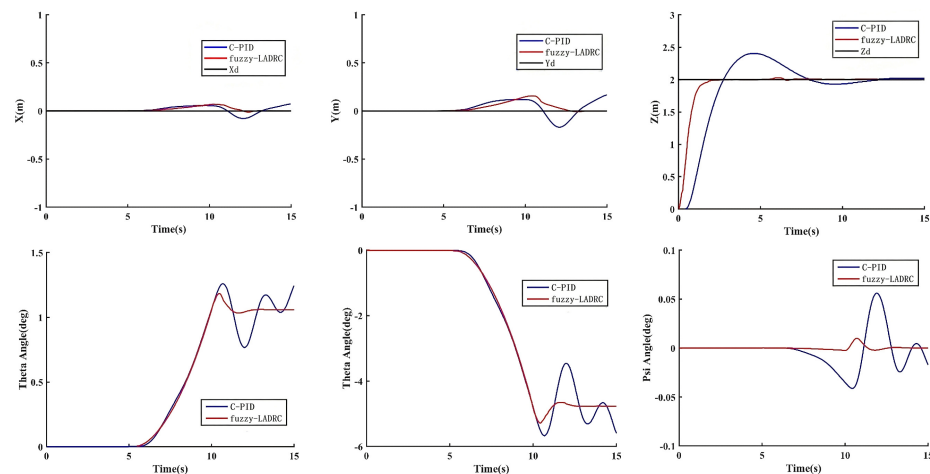


Figure 12. Six-degree-of-freedom response curves of a quadrotor UAV under a gradient wind field.

The performance comparison clearly shows that the proposed controller delivers significantly reduced oscillations across all response curves under gradient wind, outperforming the conventional cascaded PID controller. In particular, the peak fluctuations in pitch and roll angles for the proposed controller are approximately 1.2° and -5.2° , respectively; in contrast, the cascaded PID controller shows larger fluctuations of 1.3° and -5.8° , respectively. The superiority of the proposed controller is further highlighted in the yaw control channel, where it achieves near-zero oscillations, whereas the cascaded PID controller suffers from peak fluctuations of -0.06° and 0.09° at 11 s and 14 s, respectively. Furthermore, the proposed controller settles all response curves 1 s faster than the cascaded PID controller, demonstrating a markedly quicker response. This enhanced response enables the proposed controller to adapt much more swiftly to gradient wind, thus providing superior system stability. For position control, the proposed controller maintains smaller fluctuations on the XYZ axes than its counterpart, with the Z-axis channel exhibiting almost no overshoot—further evidence of the proposed controller’s stronger anti-interference capability under gradient wind conditions.

4.4.3. Simulation and Analysis of Wind Disturbance Rejection Control for Composite Wind Conditions

A stochastic wind profile with a peak wind speed of 1 m/s was applied to the entire composite wind farm. A ramp wind profile, with a peak wind speed of 8 m/s and a ramp time of 5 s, was initiated at 0 s, followed by a constant wind speed of 8 m/s after 5 s. The control performance comparison between the proposed controller and the conventional cascade PID controller is presented in Figure 13.

Under compound wind-field disturbances, the dynamic responses of both controllers become more oscillatory because of the stronger coupling and randomness of the wind excitation. Nevertheless, compared with the cascaded PID controller, the proposed controller still maintains smaller peak deviations and better overall stability, indicating stronger disturbance-rejection capability under complex wind conditions. The wind speed within the compound wind field demonstrates a trending behavior, with random increases or decreases in magnitude at subsequent time steps, thereby intensifying the disturbance. In the roll and pitch channels, the proposed controller demonstrates reduced peak values compared to the PID controller, indicating slightly superior control performance, with an approximate 5% improvement. In the yaw channel, the proposed controller exhibits peak and trough values of 0.04° and -0.03° , respectively, whereas the cascaded PID controller shows peak and trough values of 0.05° and -0.06° . This represents an approximate 30% enhancement in the yaw control channel with the proposed controller. In the XY control

channels, the proposed controller, in comparison to the cascaded PID, generates smaller displacements during the mixed phase of gradient and random winds, and exhibits reduced fluctuations during the mixed phase of constant and random winds. In the Z-axis control channel, the proposed controller demonstrates reduced overshoot during the takeoff phase, approximately 5%, compared to the cascaded PID controller, which exhibits an overshoot of approximately 20%. Furthermore, during the mixed phase of constant and random winds, the proposed controller demonstrates enhanced stability, without generating fluctuations under wind disturbance.

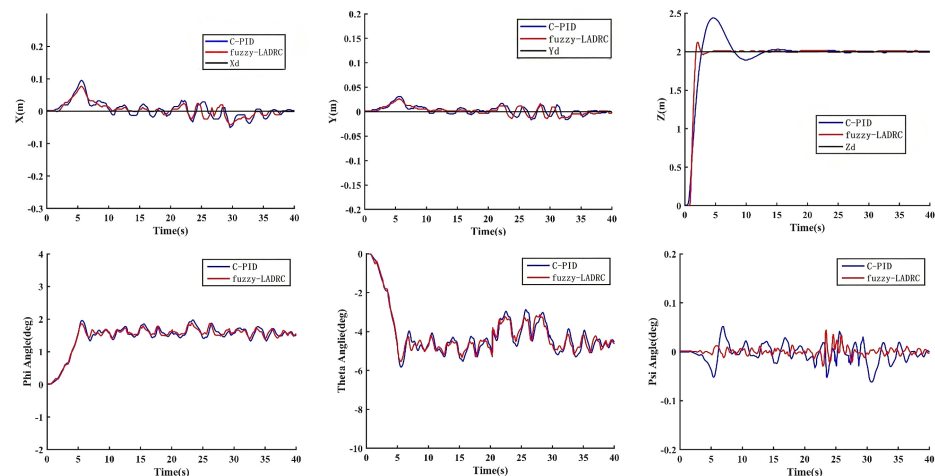


Figure 13. Six-degree-of-freedom response curves of a quadrotor UAV under compound wind conditions.

4.4.4. Quantitative Analysis of the PID Controller and the Fuzzy-LADRC Controller Under Different Wind Disturbances

To make the above comparisons more explicit, the key quantitative performance metrics extracted from the response curves under representative wind-field conditions are summarized in Table 6.

Table 6. Comparison of controller performance under representative wind-field conditions.

Wind Disturbance Type	Controller	Roll Peak Fluctuation (Deg)	Pitch Peak Fluctuation (Deg)	Yaw Peak/Trough (Deg)	Z-Axis Overshoot
Gust wind	PID	-3.7°	2.6°	-0.06 – 0.09°	With certain fluctuations
	Fuzzy-LADRC	-3°	1.8°	negligible	almost none
Gradient wind	PID	-5.8°	1.3°	-0.06 – 0.09°	With certain fluctuations
	Fuzzy-LADRC	-5.2°	1.2°	negligible	almost none
Compound wind	PID	larger	larger	-0.06 – 0.05	about 20%
	Fuzzy-LADRC	about 5% lower	about 5% lower	-0.03 – 0.04	about 5%

As shown in Table 6, the proposed controller achieves smaller peak fluctuations and better recovery performance under all three representative wind-field conditions, which further confirms its stronger disturbance-rejection capability.

5. Discussion

This paper proposes advanced methodologies to enhance the robustness and parameter-tuning efficiency of the linear active disturbance rejection controller (LADRC) under complex disturbance scenarios and validates their efficacy through simulation studies.

To address the cumbersome parameter tuning issue of LADRCs, this paper proposes an optimization method based on a hybrid PSO algorithm. Innovatively, a hybrid PSO-

GWO algorithm has been developed by integrating the position update mechanisms of PSO and GWO, significantly enhancing the algorithm's optimization capability. To validate the effectiveness of the proposed method, it is applied to LADRC parameter tuning, incorporating MATLAB code and Simulink modeling. Comparative experiments are conducted against the GWO algorithm and traditional manual parameter tuning methods. The experimental results demonstrate that the control performance achieved with parameters tuned by the hybrid PSO-GWO algorithm significantly surpasses that of the GWO algorithm and manual tuning methods, proving the superiority of the proposed method in terms of optimization efficiency and accuracy.

To address the degraded control performance of fixed-parameter controllers under complex disturbances, a parameter-adaptive Fuzzy-LADRC controller is proposed. By innovatively integrating fuzzy control theory, the fuzzy sets, quantization factors of the fuzzy universe, and membership functions are designed based on the analysis of the LADRC structure and the characteristics of the quadrotor UAV system. Furthermore, fuzzy inference rules are formulated based on expert knowledge. To evaluate the performance of the proposed controller, simulated wind fields, including gust, gradient wind, and composite wind scenarios, are constructed for simulation experiments. The results indicate that the Fuzzy-LADRC controller outperforms the traditional cascade PID controller across all simulated wind fields, demonstrating its enhanced adaptability and robustness in complex disturbance environments.

6. Limitations of the Present Study

Although the proposed controller shows promising performance under representative wind-disturbance scenarios, the validation in this study is mainly based on deterministic simulations. Statistical validation methods such as Monte Carlo simulations under stochastic wind conditions and repeated independent optimization trials were not included. Moreover, real-world flight experiments under complex wind environments were not conducted due to the difficulty of constructing repeatable wind-disturbance conditions and ensuring flight safety under strong wind scenarios. Therefore, further experimental validation and statistical robustness analysis will be considered in future work to comprehensively evaluate the practical applicability of the proposed controller.

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Data Availability Statement: The data presented in this study are available on reasonable request from the corresponding author. The data are not publicly available because the dataset is part of ongoing research and has restricted access.

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